

Eg. 1.

n and k are positive integers.

s.t. $n = 2k$

Show that $\frac{n!}{2^k} \rightarrow$ always an integer.

$n \rightarrow$ set/seq of symbols.

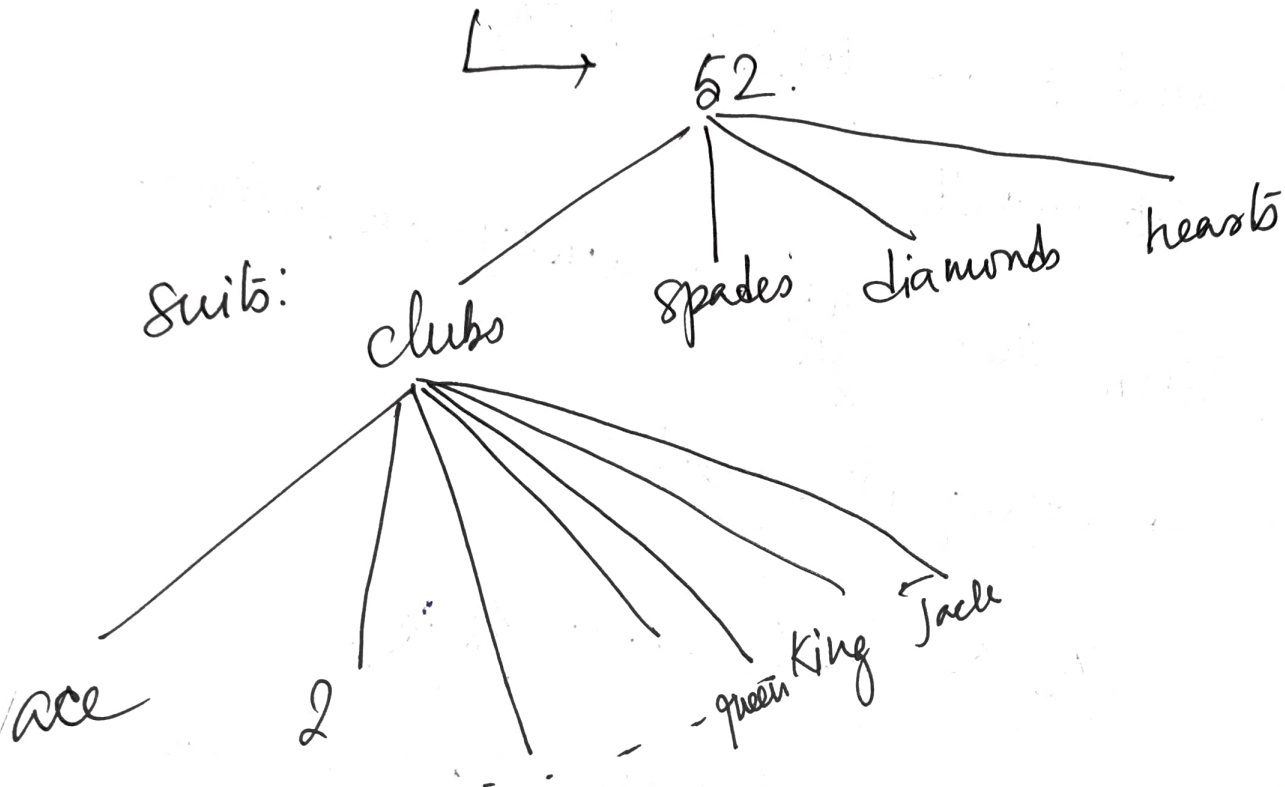
$(x_1, x_1) (x_2, x_2), \dots, (x_k, x_k)$

? What is the no. of ways that we can arrange all these n symbols ($n = 2k$)

$\hookrightarrow$ Integer

$$\frac{n!}{\underbrace{2! \cdot 2! \cdots 2!}_{k \text{ times}}} = \frac{n!}{2^k} \rightarrow \text{integer.} \quad \square$$

Cards.



Draw three cards from a standard deck.
 in succession & without replacement

$$52 \times 51 \times 50 = \frac{52!}{49!} = P(52, 3).$$

AH (ace of hearts), KD (King of diamonds)
 9C (nine of clubs).

AH - 9C - KD

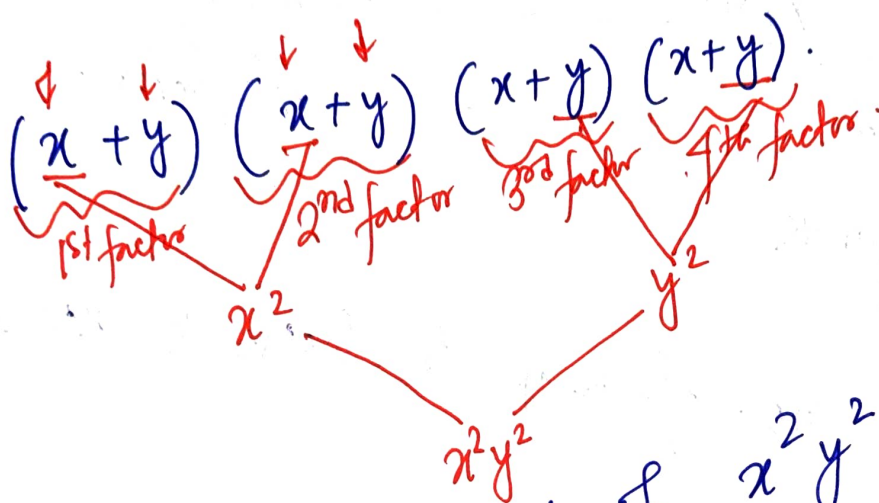
or AH - KD - 9C

or 9C - AH - KD

or KD - 9C - AH

& committee

$$n=4)$$



search of coefficient of x^2y^2

Factors selected for x		Factors selected for y	
(i)	1, 2	3, 4	6
(ii)	1, 3	2, 4	
(iii)	1, 4	2, 3	
(iv)	2, 3	1, 4	
(v)	2, 4	1, 3	
(vi)	3, 4	1, 2	

$$\hookrightarrow \binom{4}{2} = 6.$$

$(x+y)$ $(x+y)$ $(x+y)$ $(x+y)$
 1st factor 2nd factor nth factor .

$$x^k y^{n-k}$$

↳ no. of ways to choose k x 's &
 $n-k$ y 's.

$$C(n, k) = C(n, \underline{n-k})$$

E.g. 3.

(a) Coefficient of $x^5 y^2$

$$= \binom{7}{5} = \binom{7}{2} = 21$$

(b) Coefficient of $a^5 b^2$ in $(2a-3b)^7$

Assumption: $x=2a$,
 $y=-3b$.

$$\text{Coefficient} = \binom{7}{5} 2^5 (-3)^2$$

Combination of three $\rightarrow 3!$

$3! \times (\text{Number of Selection of size three from a deck of 52})$

$= \text{No. of permutation of size 3 from the deck of 52.}$

$$= P(52, 3).$$

$$\text{No. of selection of size 3} = \frac{P(52, 3)}{3!}$$

Combination

The no. of combinations of size r from a collection of size n , denoted by $C(n, r)$ or

$\binom{n}{r}$ where $0 \leq r \leq n$ satisfies

$$r! \times C(n, r) = P(n, r).$$

$$\Rightarrow C(n, r) = \frac{P(n, r)}{r!}$$

order relevant
 $\rightarrow$ permutation

order not relevant
 $\rightarrow$ combination.

Eg. 2 (a) A student taking a history exam.

Direction: any seven of 10 questions.
How many ways?

$$\binom{10}{7} = 120 \text{ ways.}$$

(b)

Answer three questions from the first five
and four questions from the rest five.

$$\binom{5}{3} \binom{5}{4} = 10 \times 5 \text{ ways.}$$

(c)

Answer seven questions from 10
these at least three are selected from
the first five.

Case 1: 3 from first 5 & 4 from the last five.

$$\binom{5}{3} \times \binom{5}{4} = 50 \text{ ways}$$

Case 2: 4 from first 5 & 3 from the last 5

$$\binom{5}{4} \binom{5}{3} = 50 \text{ ways.}$$

Case 3: 5 from the first 5 & 2 from the last 5

$$\binom{5}{5} \binom{5}{2} = 10 \text{ ways.}$$

Case 1, Case 2 & Case 3 are mutually exclusive
 $\therefore$ Total no. of ways = $50 + 50 + 10 = 110$.

Eg. 3. A gym teacher must make up four volleyball teams of nine girls from a 36 freshman girls cohort. How many ways to select these 4 teams — call the teams A, B, C, D.

Form A: $\binom{36}{9}$

Form C: $\binom{18}{9}$

Form B: $\binom{27}{9}$

Form D: $\binom{9}{9}$

No. of ways: $\binom{36}{9} \binom{27}{9} \binom{18}{9} \binom{9}{9}$.

1st student 2nd student 36th student.

A, B, C, D.

Distribute 9 A's, 9 B's, 9 C's, 9 D's

$$\frac{36!}{9! 9! 9! 9!}$$

Binomial theorem.

$$(x+y)^n = \binom{n}{0} x^0 y^n + \binom{n}{1} x^1 y^{n-1} + \dots + \binom{n}{n} x^n y^0$$

$$= \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}.$$

Proof:

Coefficient of $x_1^{n_1} x_2^{n_2} \dots x_t^{n_t}$ is the
no. of ways we can select x_1 from n_1
of the n factors, x_2 from n_2 of the
 $n - n_1$ factors, x_3 from n_3 of the
 $n - n_1 - n_2$ factors - - - - -

x_t from n_t of the
 $n - n_1 - n_2 - \dots - n_{t-1}$ factors.

$$\binom{n}{n_1} \binom{n - n_1}{n_2} \dots \binom{n - n_1 - n_2 - \dots - n_{t-1}}{n_t}$$

$$\frac{n!}{n_1! n_2! \dots n_t!}$$

$$\boxed{\binom{n}{n_1, n_2, \dots, n_t}}$$

Corollary:

$$(1) \binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n} = 2^n$$

$x=1, y=1.$

$$(2) \binom{n}{0} - \binom{n}{1} + \dots + \binom{n}{n} = 0$$

$x=1, y=-1.$

Multinomial theorem.

For any positive integers n, t the coefficient of the term $x_1^{n_1} x_2^{n_2} \dots x_t^{n_t}$ in the expansion of $(x_1 + x_2 + \dots + x_t)^n$ is

$$\frac{n!}{n_1! n_2! \dots n_t!}$$

where each n_i is an integer $0 \leq n_i \leq n$, $1 \leq i \leq t$ and $n_1 + n_2 + \dots + n_t = n$.

E.g. 4.

(a)

Find the coefficient of $x^2 y^2 z^3$
in $(x+y+z)^7$

$$= \binom{7}{2,2,3} = 210.$$

(b) find the coefficient of $a^2 b^3 c^2 d^5$

in $(a+2b-3c+2d+5)^{16}$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $a \quad b \quad c \quad d \quad 5$

$\hookrightarrow (u+v+x+y+z)^{16}$

$\downarrow$

$$\binom{16}{2,3,2,5,4}$$

$$\binom{16}{2,3,2,5,4} \times 2^2 2^3 (-3)^2 2^5 5^4$$

Combination with replacement.

E.g. 5

Seven people → restaurant

cheese-burger(c)
fish & chips(f)

haco (t)

hot dogs (h).

Does order matter?

X's & |'s
* 's & |'s.

	c	h	t	f			
①	c	c	h	h	t	t	f
②	c	c	c	c	h	t	f
③	c	c	c	c	c	c	f
④	h	t	t	f	f	f	f
⑤	t	t	t	t	t	f	f
⑥	t	t	t	t	t	t	f
⑦	f	f	f	f	f	f	f

	c	h	t	f			
①	X	X	X	X	X	X	X
②	X	X	X	X	X	X	X
③	X	X	X	X	X	X	X
④	X	X	X	X	X	X	X
⑤	X	X	X	X	X	X	X
⑥	X	X	X	X	X	X	X
⑦	X	X	X	X	X	X	X

10 symbols.
the 7 x's
& 3 |'s.

correspondence?

$$\binom{10}{7}$$

Combinations with repetition:

r size from n distinct objects
with replacement

① n distinct object + ② $r-1$ sticks

$$\therefore \text{The choices} = \binom{n+r-1}{r}$$

for $i = 1$ to n

$j = 1$ to i

$k = 1$ to j

print ("hello").

n items.

triple. $\langle i, j, k \rangle \rightarrow$ with replacement

Catalan Numbers.

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

→ Counting problem
of balanced structures.
(non-crossing)

n	0	1	2	3	4	...
C _n	1	1	2	5	14	

then
numbers

Mountain range example.

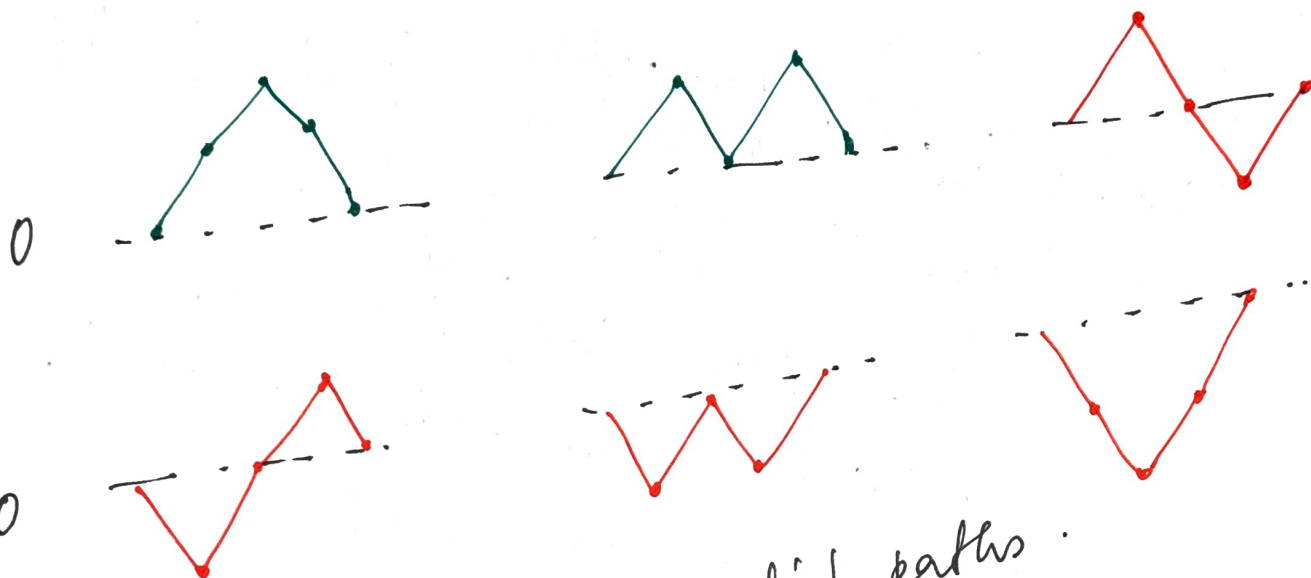
- Take a set of n ($/$) up-slashes.
- $\&$ n ($\backslash$) down-slashes
- read a path from left to right. starting from level 0 (Sea-level).
- height rises ^{1 unit} with every $/$
- height falls 1 unit with every $\backslash$

We are interested in paths that stay at or above the sea-level.

→ Dyck paths → counting them
Catalan numbers

E.g 6.

2 ups & 2 downs.



$$\frac{1}{3} \binom{4}{2} = 2 \text{ valid paths.}$$

Step 1:

Count everything:

$$\binom{2n}{n}$$

Step 2:

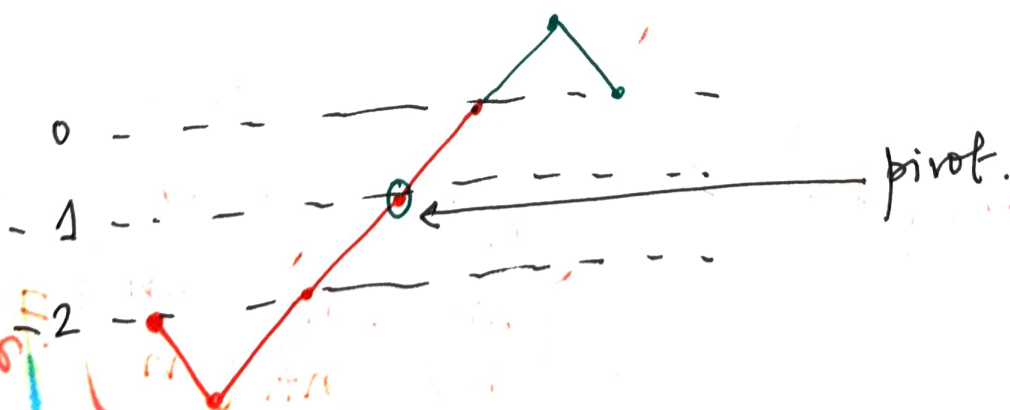
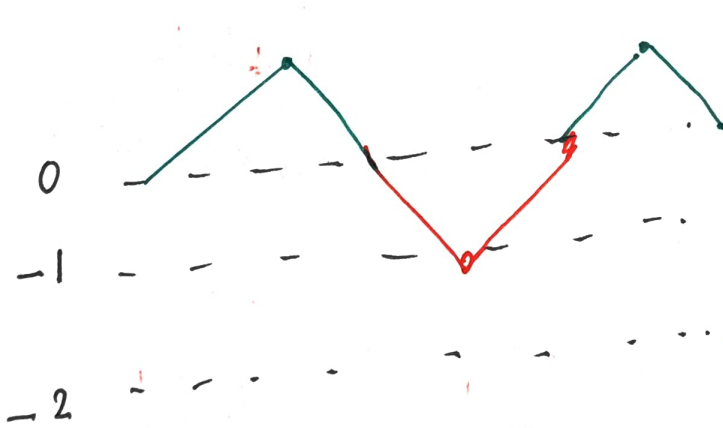
We have to account for bad paths.
→ reflection principle.

① Take a bad path & find the first step where the height reaches -1 .

② Flip every step before that step (including the first dip)

③ Rest remains as-is.

$\therefore$ Path started from 0 . & remained so till it reached -1 . flipping changes the starting ~~height~~ height from $0 \rightarrow -2$.



bijection

Counting is done from 1 to

bad paths $(0,0) \rightarrow (2n,0) \xleftrightarrow{\text{reflect}} (0,-2) \rightarrow (2n,0)$

A path from $(0,-2)$ to $(2n,0)$ is a sequence of $\uparrow$ & $\downarrow$

(u) up-steps - down-steps(d) = 2
 end-height - start height = $0 - (-2) = 2$

$$\begin{aligned} u - d &= 2 \\ u + d &= 2n \end{aligned}$$

$$\begin{aligned} u &= n+1 & \rightarrow & \text{force} \\ d &= n-1 & \rightarrow & \text{up-steps} \\ & & & \text{down-steps} \end{aligned}$$

bad paths : $\binom{2n}{n+1}$ or $\binom{2n}{n-1}$

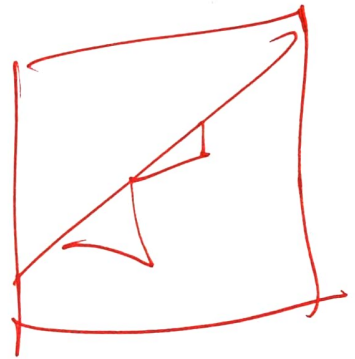
"Good paths" = $\binom{2n}{n} - \binom{2n}{n-1}$
 $= \binom{2n}{n} - \frac{n}{n+1} \binom{2n}{n} = \frac{1}{n+1} \binom{2n}{n}$

combinatorics
 2n
 n
 n+1

(E.g. 1) (/) \ → balanced parenthesis.

(E.g. 2) valid sequences of push & pops.

(E.g. 3) Full binary tree
both n in internal
nodes.



(n=3)

